relative brightness of those colors. It incorrectly predicted single colors where pairs or triplets of closely spaced colors appeared instead. And finally, atomic spectra were susceptible to external influences, some of which couldn't be fully accounted for by Bohr's model. Put an atom in a magnetic field, and its spectrum changed. Put it in an electric field, and its spectrum changed in a different way. Colors shifted, blurred, and split, dimmed and brightened, with no larger pattern in sight—until Heisenberg.

In June 1925, Heisenberg came down with a hideous case of hay fever. Sneezing and nearly blind, with tears streaming down his impressively swollen face, the desperate young physicist took two weeks' vacation to the island of Heligoland, a small barren island in the North Sea, utterly devoid of trees and flowers. After several days on the island, he recovered and resumed his research. Ignoring everything Bohr's model said about the orbits of electrons in atoms, Heisenberg focused on what he could actually see: the spectrum of light emitted from the jumps between energy levels themselves. Working alone at three in the morning, in a shack on a rock battered by a frigid sea, his hands shaking, excitedly fumbling over "countless arithmetical errors," Heisenberg made a breakthrough. "I had the feeling that, through the surface of atomic phenomena, I was looking at a strangely beautiful interior, and felt almost giddy at the thought that I now had to probe this wealth of mathematical structures nature had so generously spread out before me." Heisenberg had developed a strange new mathematics on the fly, one in which simple statements like "three times two equals two times three" were not always true. Using this unwieldy math, Heisenberg had found a way to predict the spectrum of a quantum oscillator—a tiny pendulum—which, in turn, allowed him to predict how atomic spectra respond to magnetic fields.

When Heisenberg returned to his job at the University of Göttingen, he cautiously sent a draft of his new theory to his friend, the brilliant physicist Wolfgang Pauli—"generally my severest critic," as Heisenberg recollected years later—who greeted the new theory with effusive praise. "[Heisenberg's ideas offer] a new hope, and a renewed enjoyment of life.… Although it is not the solution to the riddle, I believe that it is now once again possible to move forward," said Pauli. Max Born, Heisenberg's supervisor, agreed. Born and his student Pascual Jordan helped Heisenberg to elucidate the structure and implications of his new theory, which Born